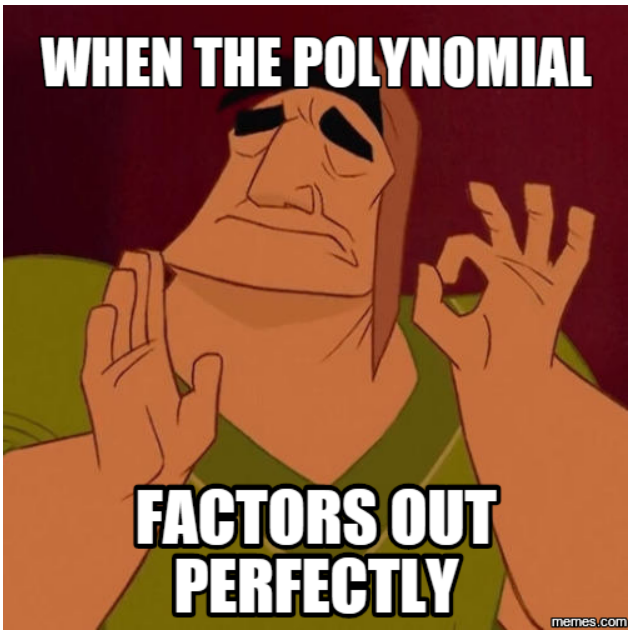

Grade 12 Advanced Functions
Polynomial Functions Test

First Name: _____

Last Name: _____

Directions:

- Please answer to 2 decimal points
- The test is designed to be completed in 75 minutes

For grading use only

Page:	2	3	4	Total
Points:	8	32	20	60
Score:				

Multiple Choice (10 marks)

1. (1 point) The minimum number of turning points that a quartic function can have is:
   A. 0
   B. 1
   C. 2
   D. 3
2. (1 point) The degree of the polynomial $f(x) = -x^4 + 5x^3 - 6x^2 + 3$ is:
   A. 2
   B. 3
   C. 4
   D. 5
3. (1 point) The leading coefficient of $f(x) = -x^4 + 5x^3 - 6x^2 + 3$ is:
   A. -6
   B. 1
   C. 5
   D. -1
4. (1 point) Which of the following is NOT a factor of $x^3 - 5x^2 + 8x - 40$?
   A. $x - 5$
   B. $x + 4$
   C. $3x - 2$
   D. $2x - 3$
5. (1 point) The end behavior of $f(x) = -5x^4 + 3x^3 - 2x^2 + 6x - 9$ is:
   A. $x \to -\infty, f(x) \to -\infty$; $x \to \infty, f(x) \to -\infty$
   B. $x \to -\infty, f(x) \to \infty$; $x \to \infty, f(x) \to \infty$
   C. $x \to -\infty, f(x) \to \infty$; $x \to \infty, f(x) \to -\infty$
   D. $x \to -\infty, f(x) \to -\infty$; $x \to \infty, f(x) \to -\infty$
6. (1 point) When $f(x) = 3x^3 + 5x^2 - 2x + 8$ is divided by $x + 2$, the remainder is:
   A. 8
   B. 0
   C. -2
   D. 30
7. (1 point) Divide $(6x^4 + 7x^3 - 21x^2 + 9)$ by $(x - 3)$. What is the remainder?
   A. 0
   B. 9
   C. -6
   D. 12
8. (1 point) Factor $3x^4 - 8x^3 + 13x^2 - 30x$ completely. Which is NOT a factor?

A. x
B. $x - 2$
C. $3x - 5$
D. $x + 2$

9. (1 point) The polynomial $f(x) = x^5 + 2x^3 - x$ has how many x-intercepts?
   A. 1
   B. 2
   C. 3
   D. 5

10. (1 point) If $f(x) = x^4 - 4x^2$, which of the following is a turning point?
   A. $(0, 0)$
   B. $(2, 0)$
   C. $(\sqrt{2}, -4)$
   D. $(-2, 0)$

Long Answer (40 marks)

11. Polynomial Division
   (a) (2 points) Divide $f(x) = x^3 - 4x^2 + 5x - 2$ by $x - 1$ and state the remainder.
   (b) (2 points) Write the quotient and remainder in the form $f(x) = (x - 1)Q(x) + R$.

12. Equation from Graph
   (a) (2 points) A cubic function passes through $(0, 0)$, $(2, 0)$, and $(4, 0)$, and has a y-intercept of 0. Write its equation.
   (b) (2 points) Sketch the graph, labeling all intercepts.

13. Factoring Polynomials
   (a) (2 points) Factor $x^4 - 5x^3 + 6x^2$ completely.
   (b) (2 points) List all real zeros and their multiplicities.
   (c) (2 points) Sketch the graph, labeling all intercepts and describing the behavior at each zero.

14. Factoring with Rational Root Theorem
   (a) (2 points) Factor $2x^3 - 7x^2 + 6x$ completely.
   (b) (2 points) List all real zeros and their multiplicities.
   (c) (2 points) For each zero, describe how the graph behaves at that point.

15. Factoring Higher Degree
   (a) (2 points) Factor $x^5 - 3x^4 - 4x^3$ completely.
   (b) (2 points) List all real zeros and their multiplicities.
   (c) (2 points) Sketch the graph, labeling all intercepts and describing the behavior at each zero.

16. Graph Analysis
   (a) (2 points) Factor $f(x) = x^3 - 6x^2 + 11x - 6$ and list all zeros.
   (b) (2 points) Find the coordinates of all turning points.

(c) (2 points) Sketch the graph, labeling all intercepts and turning points.

(d) (2 points) State the intervals where the function is increasing and decreasing.

17. Graph from Factored Form

(a) (2 points) Given $f(x) = (x+2)^2(x-1)$, list all zeros and their multiplicities.

(b) (2 points) Describe the behavior of the graph at each zero.

(c) (2 points) State the end behavior.

(d) (2 points) Sketch the graph, labeling all intercepts and describing the behavior at each zero.

18. Graph with Triple Root

(a) (2 points) Factor $f(x) = x^3 - 3x^2 + 3x - 1$ and list all zeros and their multiplicities.

(b) (2 points) Describe the behavior of the graph at the root.

(c) (2 points) Find the y-intercept.

(d) (2 points) Sketch the graph, labeling all intercepts and describing the behavior at the root.